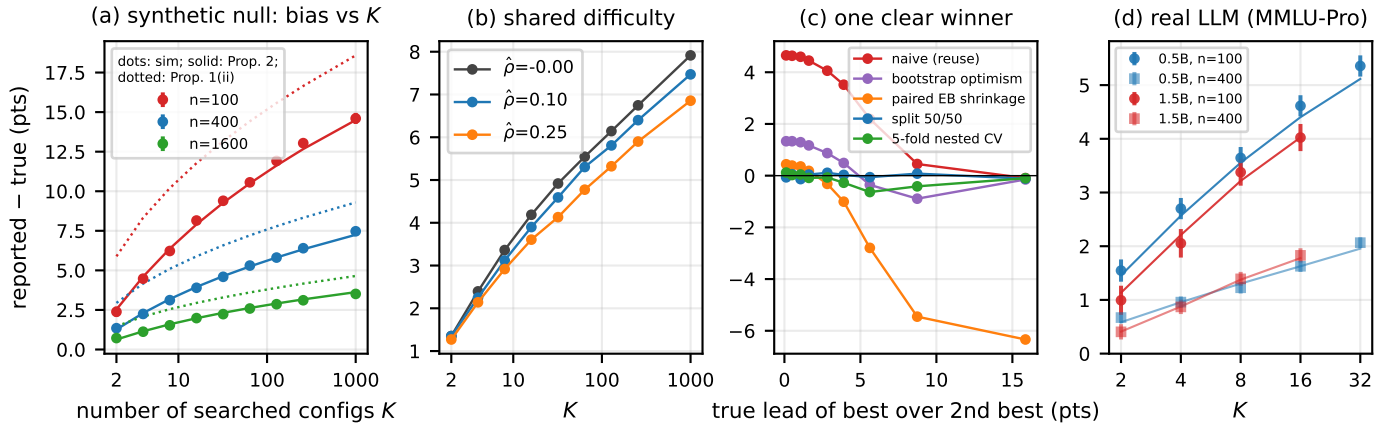
(a) synthetic null: bias vs K
dots: sim; solid: Prop. 2;
dotted: Prop. 1(ii)
n=100
n=400
n=1600
reported − true (pts)
number of searched configs K
(b) shared difficulty
$\hat{\rho}$=-0.00
$\hat{\rho}$=0.10
$\hat{\rho}$=0.25
K
(c) one clear winner
naive (reuse)
bootstrap optimism
paired EB shrinkage
split 50/50
5-fold nested CV
true lead of best over 2nd best (pts)
(d) real LLM (MMLU-Pro)
0.5B, n=100
0.5B, n=400
1.5B, n=100
1.5B, n=400
K